MissionEV

A Strava for carbon. Reward the impact of the journey, not the ownership of the vehicle.

Live on Portal V3 testnet

`0x0904244017324dc1bf7c611fe1236936ee65fc82`

Chain 420420777 · deployed and verified

Open source, MIT

`github.com/ManavNegi2807/missionev-app`

`Live demo: missionev.vercel.app`

Next.js · Leaflet + OpenStreetMap · Tesseract.js · viem

THE PROBLEM

Incentives reward owning a clean vehicle, not making a clean journey

- A commuter who takes the metro every day earns nothing, while an EV owner who takes a petrol cab for a day is treated as clean.
- There is no cheap, verifiable record of what a single journey actually saved.
- Without that record, an employer, a city or a sponsor cannot pay for behaviour change with confidence.

THE IDEA

Measure every journey. Reward only what can be verified.

1 CHOOSE
How you travelled for this trip

2 RECORD
Real GPS, or a labelled simulation

3 MEASURE
Compare with a stated reference journey

4 VERIFY
The verifier node recomputes and signs

5 CLAIM
Your own wallet submits; $MEV is minted

Metro, 1.5 kg avoided

Day 1: 1.5 × 20 = 30 MEV + 10 bonus = **40 MEV**

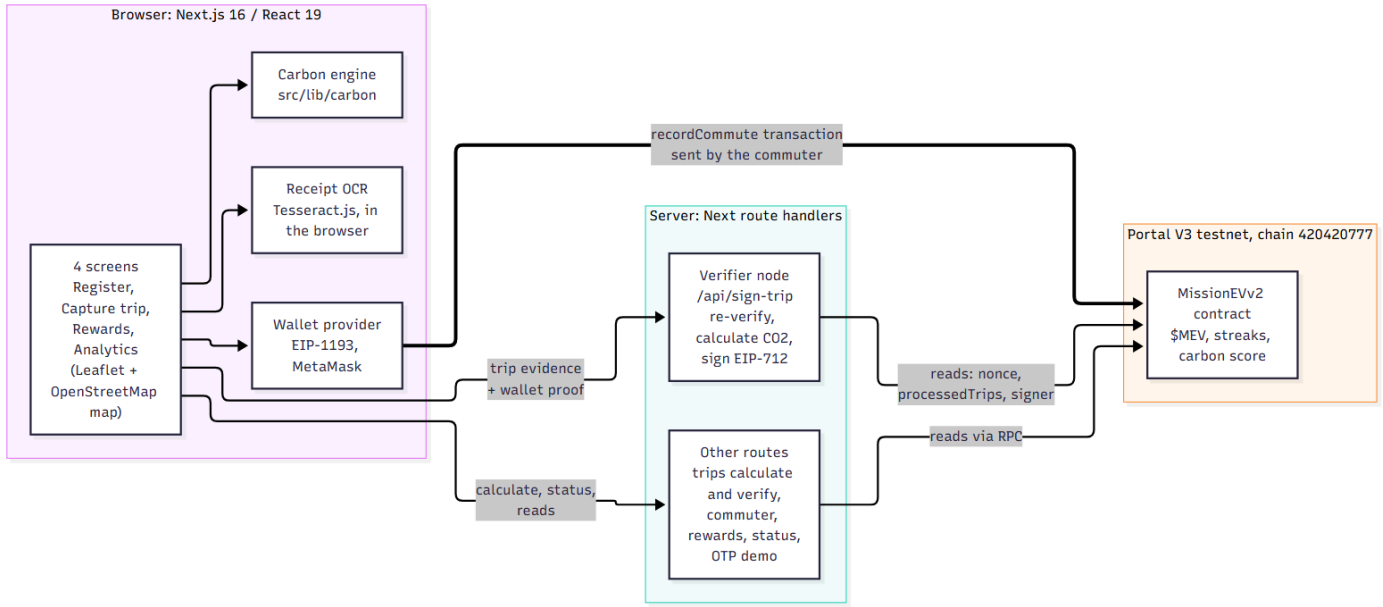
Petrol cab

Measured and shown, **never rewarded**, even if you own an EV

An EV is not zero

Electricity counts at the grid's emission factor (CEA, FY 2025-26)

The server signs; the commuter's wallet submits



- The verifier node recomputes distance and carbon itself and signs **only what it derived**.
- The contract pays `msg.sender`, so the oracle can sign a trip but **can never submit it**.
- Streak and carbon score live on-chain.
- Petrol, diesel and simulated trips are never signed.

EVM compatibility is what makes this practical

- **Standard tools just work:** EIP-712 typed-data signatures, `ecrecover` and an ERC-20 token, with ordinary wallets.
- **Commuter-submitted:** users keep control; the oracle needs no funds and no custody.
- **An open record:** balances, streaks, score and processed trips are public.
- **Honest scope:** we use the EVM layer. We do not claim WASM or cross-chain features.

Deployed and verified

Contract
`0x0904244017324dc1bf7c611fe1236936ee65fc82`

Deploy transaction (block 309126)
`0x11562d9b34bc9236413a1d49a7712179d8f50defdc610b3c81c6dbf01d9ca965`

Check it yourself
`npm run contract:verify`

PROOF IT WORKS

Built, tested and checked against the live chain

5/5

Deployment checks: code, EIP-712 domain, token, oracle and owner all match the artifact

Live

The deployed contract accepted the verifier's signature in a simulation; a tampered copy and another commuter were rejected

120+

Tests, including the compiled contract run in an EVM; CI green; a fresh clone installs and builds

3 flows

Browser-tested: full demo 41/41, on-chain flow on a local chain 15/15, receipt OCR 16/16

Open

Open source by default: Leaflet + OpenStreetMap, in-browser Tesseract.js, Next.js, viem. A Google Maps key is an optional override

Next

The first real `recordCommute` from the app needs about 35 tPOT of testnet gas; that is the one gap we are closing

BUSINESS POTENTIAL

Who pays, and how we will test it

Employers and campuses

A programme fee per active commuter, with an emissions report

Operators and cities

Funded reward pools and evidence of mode shift on a corridor

Sponsors

Reward pools tied to a measured outcome, not an impression

- **First users:** one corridor or campus in Delhi NCR, where the grid factor and transit modes are already modelled.
- **Before a public pilot:** real phone verification, a database for the oracle's limits, sourced metro and e-bus factors.
- **We name our risks:** GPS spoofing, $MEV is not a carbon credit, factor quality, a single oracle key, transaction cost.

Reusable for every Portal V3 builder

What we give back

- MIT-licensed code and a worked EIP-712 oracle pattern
- Dry-run deploy and an immutable-aware contract verifier
- `npm run network:health`: compares the EVM endpoint with the native node. It would have caught the two-day RPC freeze we hit
- Builder notes on fees, the Remix error and RPC lag

What would help us most

- Gas support: about 418 tPOT to deploy and about 35 tPOT per call today
- A technical review of the oracle and its security boundaries
- Mainnet migration guidance
- Introductions for a transit or employer pilot

Next, in order

A first real on-chain trip from the app, then real phone sign-in and a database for the oracle's limits, sourced metro and e-bus factors, and one corridor or campus pilot in Delhi NCR.

Reward the journey.

Live demo

missionev.vercel.app

Code

github.com/ManavNegi2807/missionev-app

Explorer

node-console.portaldot.dev

```
Contract 0x0904244017324dc1bf7c611fe1236936ee65fc82
Deploy tx 0x11562d9b34bc9236413a1d49a7712179d8f50defdc610b3c81c6dbf01d9ca965
```

Team: Manav Negi (@ManavNegi2807) and teammates. $MEV is a prototype reward token, not a certified carbon credit.